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Lingzhi Wang

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Appointments

University of South Carolina — Columbia, SC
Incoming Tenure-Track Assistant Professor — Starting August 2026
Department of Integrated Information Technology

Education

Northwestern University — Evanston, United States
Ph.D Computer Science (GPA: 3.98/4.0) — May 2021 – June 2026
Advisor: Prof. Yan Chen

Tsinghua University — Beijing, China
B.E Electronic Information Science and Technology (GPA: 3.6/4.0) — Aug 2016 – June 2020
B.Ec Economics (double major) (GPA: 3.8/4.0) — May 2017 – June 2020

Coursework

Courses: Object-Oriented Programming, Data Structures & Algorithms, Embedded Systems, Discrete Math, Linear Algebra, Calculus, Physics, Probability & Statistics, Machine Learning, Artificial Intelligence

Research Interest

My research focuses on applying advanced machine learning and artificial intelligence techniques, particularly LLMs, to strengthen operational cyber defense. Representative questions I work on include: 1) how cutting-edge AI methods can be integrated into practical security and network systems; 2) how to model human knowledge and reasoning with LLMs to carry out complex cybersecurity tasks; 3) how to build higher-quality research infrastructure for cybersecurity, including datasets, benchmarks, testbeds, and knowledge bases; and 4) how to identify, assess, and mitigate the emerging security risks introduced by LLMs and autonomous agents.

Publications

Note: * indicates co-first authorship.

1. Cai, Q.*, **Wang, L.***, Zhu, Y., Chen, Z., Shen, X., & Li, Z. Building Next-Generation Datasets for Provenance-Based Intrusion Detection. *Presented at Workshop on Attack Provenance, Reasoning, and Investigation for Security in the Monitored Environment.* (PRISM @ NDSS'26). [paper]

2. **Wang, L.**, Shi, X., Li, Z., Jiang, Y., Tan, S., Jiang, Y., Cheng, J., Cheng, W., Shen, X., Li, Z., & Chen, Y. Automated Penetration Testing with LLM Agents and Classical Planning. *Arxiv preprint, 2025.* [arxiv]

3. **Wang, L.**, Yegneswaran, V., Shi, X., Li, Z., Gehani, A., & Chen, Y. GraphFaaS: Serverless GNN Inference for Burst-Resilient, Real-Time Intrusion Detection. Presented at *Workshop on ML for Systems at NeurIPS 2025.* [paper]

4. **Wang, L.**, Li, Z., Jiang, Y., Wang, Z., Shen, X., Ruan, W., & Chen, Y. From Sands to Mansions: Actionable, Customizable, and Causality-Preserving Cyberattack Emulation with LLM-powered Symbolic Planning. Accepted by *the 24th International Conference on Applied Cryptography and Network Security* (ACNS 2026). [arxiv]

5. Li, Z.*, **Wang, L.***, Wang, Z.*, Shen, X., Xu, H., Chen, Y., & Ji, S. Incorporating Gradients to Rules: Towards Online, Adaptive Provenance-based Intrusion Detection. In *IEEE Transactions on Dependable and Secure Computing* (IEEE TDSC). [paper]

6. Shen, X.*, **Wang, L.***, Li, Z., Chen, Y., Zhao, W., Sun, D., Wang, J., & Ruan, W. PentestAgent: Incorporating LLM Agents to Automated Penetration Testing. In *Proceedings of the 20th ACM Asia Conference on Computer and Communications Security* (AsiaCCS'25) [paper][code]

7. **Wang, L.***, Shen, X., Li, W., Li, Z., Sekar, R., Liu, H., & Chen, Y. Incorporating Gradients to Rules: Towards Lightweight, Adaptive Provenance-based Intrusion Detection. In *Proceedings of the 32nd Network and Distributed System Security Symposium 2025* (NDSS'25) [paper][code]

8. Shen, X., Li, Z., Burleigh, G., **Wang, L.**, Chen, Y. Decoding the MITRE Engenuity ATT&CK Enterprise Evaluation: An Analysis of EDR Performance in Real-World Environments. In *Proceedings of the 19th ACM Asia Conference on Computer and Communications Security* (AsiaCCS'24). [paper][code]

9. Wang, J.*, **Wang, L.***, Yu, H., Shen, X., & Chen, Y. Paris: A Practical, Adaptive Trace-Fetching and Real-Time Malicious Behavior Detection System. *Preprint* arXiv:2411.01273. [arxiv]

10. **Wang, L.**, Zhao, N., Chen, J., Li, P., Zhang, W., Sui, K. Root-cause metric location for microservice systems via log anomaly detection. *2020 IEEE international conference on web services* (ICWS'20). IEEE, 2020. [paper]

Tutorials

1. **Wang, L.** Using Aurora, an automated attack emulation system, to create benchmark datasets for intrusion detection.
   Tutorial at *Workshop on Attack Provenance, Reasoning, and Investigation for Security in the Monitored Environment* (PRISM), co-located with NDSS'26.

Papers Under Review

1. Li, Z., Wei, Y., Shen, X., **Wang, L.**, Chen, Y., Xu, H., ... & Zhang, F. "Marlin: Knowledge-Driven Analysis of Provenance Graphs for Efficient and Robust Detection of Cyber Attacks" [arxiv]

2. Shen, X., Cheng, W., Chen, Y., Li, Z., Gu, Y., **Wang, L.**, Zhao, W., Sun, D., Wang, J., & Ning, Y. "AEAS: Actionable Exploit Assessment System"

Ongoing Projects

Automated Penetration Testing using LLM | *Northwestern University* Nov. 2025 – Present

- Uses a large language model for unstructured cyberattack knowledge summarization and planning problem definition.
- Addresses the challenges of using traditional AI planning and the Planning Domain Definition Language (PDDL) in penetration testing, such as managing uncertainty, handling partial observability, etc.
- Reduces human labor and domain expertise needed in manual penetration testing.

Next-generation Provenance-based Intrusion Detection System | *Northwestern University* Dec. 2025 – Present

- Addresses challenges of generating adaptive and fine-grained rules in existing provenance-based host intrusion detection systems using generative AI.
- Reduces the overhead running the real-time detection and builds a more accurate detector using less training data.
- Incorporates new data log sources (e.g. call stack traces) to mitigate novel threats such as the fileless attack.

Talks

1. **Tutorial**: *"Using Aurora, an Automated Attack Emulation System, to Create Benchmark Datasets for Intrusion Detection"*, PRISM'26 (co-located with NDSS'26), San Diego, CA, February 23, 2026.
2. **Workshop Talk**: *"Building Next-Generation Datasets for Provenance-Based Intrusion Detection"*, PRISM'26 (co-located with NDSS'26), San Diego, CA, February 23, 2026.
3. **Invited Talk**: *"From Sands to Mansions: Actionable, Customizable, and Causality-Preserving Cyberattack Emulation with LLM-powered Symbolic Planning"*, Smart Information Flow Technologies (SIFT), Online, August 13, 2025.
4. **Conference Talk**: *"Incorporating Gradients to Rules: Towards Lightweight, Adaptive Provenance-based Intrusion Detection"*, NDSS Symposium, San Diego, CA, February 25, 2025.

Experience

Obsidian Security | *Research Intern* Mar. 2026 – Jun. 2026
Researched and developed security monitoring and governance frameworks for multi-agent AI systems, focusing on permission management, access control, and threat detection in autonomous SaaS environments.

USC Information Sciences Institute (USC-ISI) | *Research Intern* Jan. 2026 – Mar. 2026
Enhanced SPHERE's software provisioning capabilities by integrating external tools into the testbed, enabling users to easily define and automatically deploy software environments.

SRI International (SRI) | *Research Intern* Jun. 2025 – Aug. 2025
Developed machine learning models for serverless computing environments, focusing on performance optimization and graph analytics for security applications

Northwestern University | *Teaching Assistant* Sep. 2021 – Jun. 2025
CS450: Internet Security, CS396/CS326: Intro to the Data Science Pipeline, DE200: Foundations of Data Science, CS217: Data Management and Information Processing, CS212: Mathematical Foundation to Computer Science

Tencent | *Research Intern* May 2020 – Aug. 2020
Work on data mining using the map data from users' smartphones.

Bizseers | *Research Intern* Mar. 2019 – Jan. 2020
Proposed a novel framework of failure localization and root-cause diagnosis using system logs and key performance indicators collected from large distributed systems, such as servers and databases in banks or E-commerce companies.

Tsinghua University | *Research Assistant* Nov. 2018 – Jun. 2019
Constructed a weighted graph of app behavior based on deep learning and graphic modeling to analyze app-using data collected by a software company in Finland, including app usage data from more than 8,000 users in 7 years, with over 20 million records.

Funding & Research Support

Google Cloud Research Credits Program, $6,000 (proposal lead; awarded to PI Prof. Yan Chen), 2025.

Services

Program Committee: Workshop on Attack Provenance, Reasoning, and Investigation for Security in Monitored Environments (PRISM), co-located with NDSS'26

Reviewer: IEEE Transactions on Dependable and Secure Computing (TDSC), IEEE Transactions on Information Forensics and Security (T-IFS), IEEE Security & Privacy, ACM Transactions on Privacy and Security (TOPS)

Artifact Committee Member: USENIX Security 2025, ACM CCS 2025

External Reviewer: NDSS 2026, IEEE S&P 2025, IEEE S&P 2024